

SALOMON ADRIAN BROU

Backend Software Engineer | Python | FastAPI | AWS | Terraform
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PROFESSIONAL SUMMARY

Backend engineer with 7+ years building and shipping production Python services and cloud infrastructure. Designs async FastAPI APIs, models and optimizes PostgreSQL, and deploys to AWS (Lambda, API Gateway, RDS, ECS Fargate, VPC, IAM) with Terraform and GitHub Actions CI/CD. Owns systems end to end: schema design, auth boundaries, serverless functions, LLM integration, and the test suites and observability that keep them alive in production. Comfortable working solo without a platform team behind the work.

TECHNICAL SKILLS

- **Languages:** Python (3.12), Go, TypeScript / JavaScript, SQL, C, Bash
- **Backend & APIs:** FastAPI (async / await), REST API design, JWT / RBAC, custom authorizers, SQLAlchemy ORM, Alembic, Pydantic, WebSockets, Redis
- **AWS:** Lambda, API Gateway, RDS PostgreSQL, DynamoDB, VPC, IAM (least-privilege), S3, CloudFront, ALB, Secrets Manager, CloudWatch, ECS Fargate, ECR
- **Infrastructure as Code & DevOps:** Terraform (modules, state management), Docker, GitHub Actions CI/CD
- **Databases:** PostgreSQL (JSONB, CTEs, window functions, query optimization), DynamoDB, Supabase (RLS, Edge Functions, Realtime), PostGIS
- **Testing & Practices:** pytest (TDD, coverage), SOLID principles, clean code, code review, structured logging, Agile
- **LLM Integration:** Anthropic Claude API, serverless AI features (AWS Bedrock runs Claude models)

EXPERIENCE

Backend Developer (Contract) — Juice Labz LLC, Streak Performance Dec 2025 to Present
Remote

- Migrating a golf fitness web application to a cross-platform mobile product: Go backend services, React Native (Expo) client.
- Working directly with the founder on architecture decisions, API design, feature scope, and delivery milestones.

Software Engineer, Backend & Platform — Anytime Fitness (TrainerOS) 2022 to Present
Huntersville, NC

- Designed, built, and operate TrainerOS (aftrainer.app), a production gym management platform serving 100+ daily users. Sole engineer across backend, data layer, AI features, and deployment.
- Built the full PostgreSQL backend: 12-table schema, 25 migrations, row-level security policies, security-definer functions, real-time subscriptions, and private file storage with signed URLs.
- Shipped 7 production AI features on serverless functions backed by the Anthropic Claude API, including workout generation, coaching notes, PR prediction, nutrition planning, and weekly recaps.
- Engineered real-time chat, Web Push notifications (VAPID), self-registration via database triggers, and CSV / PDF export pipelines.
- Optimized PostgreSQL with CTEs and window functions for sub-second responses across 1,500+ member records; automated monthly reporting that eliminated 20+ hours of manual work.
- Built and deployed a Random Forest churn-prediction pipeline (82% accuracy) flagging at-risk members 30 days early, supporting retention programs that improved outcomes by 30%.

Full Stack Developer (Freelance) — Independent 2018 to 2021
Côte d'Ivoire

- Delivered end-to-end web applications for small business clients with Python, Flask, and SQL, including a full restaurant platform with menu management, reservations, and customer-facing UI.

- Built REST APIs and relational schemas, gathered requirements directly from clients, and translated them into shipped features.

SELECTED PROJECTS

SaaS Gym CRM Backend (AWS) — FastAPI, PostgreSQL, SQLAlchemy, Alembic, Redis, Docker, AWS (ECS Fargate, RDS, ECR, Secrets Manager), GitHub Actions, pytest

- Multi-tenant CRM exposing 35+ REST endpoints with JWT authentication and role-based access control. Redis caching, SQLAlchemy ORM with Alembic migrations, and 52 passing unit and integration tests.
- Containerized with Docker and deployed to AWS ECS Fargate with RDS PostgreSQL, full CI/CD through GitHub Actions, and secrets managed in AWS Secrets Manager. Built to production tradeoffs, not just working state.

Spendless, Personal Finance API — FastAPI, PostgreSQL, SQLite, React

- Layered backend (Route, Service, Repository, Database) in a monorepo, with a deliberate SQLite-to-PostgreSQL migration to exercise schema and data-migration tradeoffs.

Jarvis, AI Coding Agent — Python, Gemini / OpenAI APIs, Strategy Pattern

- CLI agent that loops over file reads, writes, and sandboxed execution. Strategy Pattern lets it swap LLM backends; strict path validation and structured logging keep it inside its working directory.

EDUCATION & CERTIFICATIONS

- **Master's in Software Engineering** — Groupe École Supérieure de Commerce, Côte d'Ivoire (2020). Bachelor's in Software Engineering (2017).
- **Associate in Arts** — Central Piedmont Community College, Charlotte, NC (in progress, expected 2026).
- **Boot.dev** — completed Python, Backend Development, AWS Infrastructure & Cloud Computing, Linux CLI, and Git / GitHub; currently on the Go track.
- **Certifications:** Oracle SQL Developer; Boot.dev Python Developer, Backend Developer Path, AWS Infrastructure & Cloud Computing.
- **Languages:** English (Professional), French (Native).